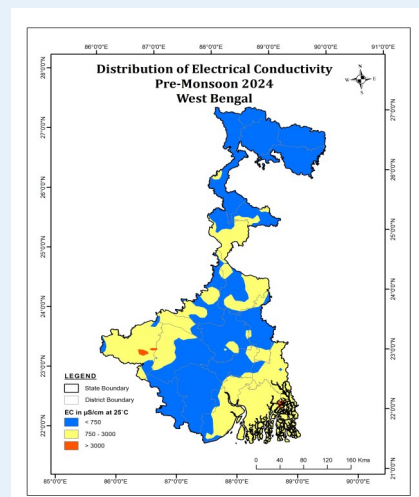
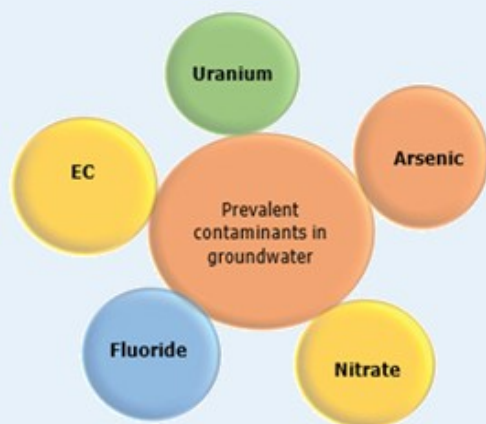


Groundwater Quality Scenario in West Bengal

Parameters	No of samples	Permissible limit	No. of Samples above permissible limit	% Samples above permissible Limit
EC	875	3000 $\mu\text{S/cm}$	13	1.49
Fluoride	875	1.5 mg/L	8	0.91
Nitrate	875	45 mg/L	78	8.91
Arsenic	236	10 ppb	46	19.49
Uranium	235	30 ppb	0	0.00



Districts with anomalous values at sporadic locations

EC (3000 $\mu\text{S/cm}$)	Bankura, Howrah, Malda, North 24 Parganas, Purulia, South 24 Parganas
Fluoride ($F > 1.5$ mg/L)	Birbhum, Dakshin Dinajpur, Malda, North 24 Parganas, Paschim Bardhaman
Nitrate (Nitrate > 45 mg/L)	Alipurduar, Bankura, Birbhum, Cooch Behar, Dakshin Dinajpur, Darjeeling, Hooghly, Howrah, Jalpaiguri, Jhargram, Malda, Murshidabad, Nadia, North 24 Parganas, Paschim Bardhaman, Paschim Medinipur, Purba Bardhaman, Purulia, South 24 Parganas
Arsenic ($As > 10$ ppb)	Coochbehar, Malda, Murshidabad, Nadia, North 24 Parganas, South 24 Parganas
Uranium ($U > 30$ ppb)	Not Any

For Further Information, Contact to :

Chairman, CGWB, Bhujal Bhawan,
NH IV Faridabad, Haryana - 121001



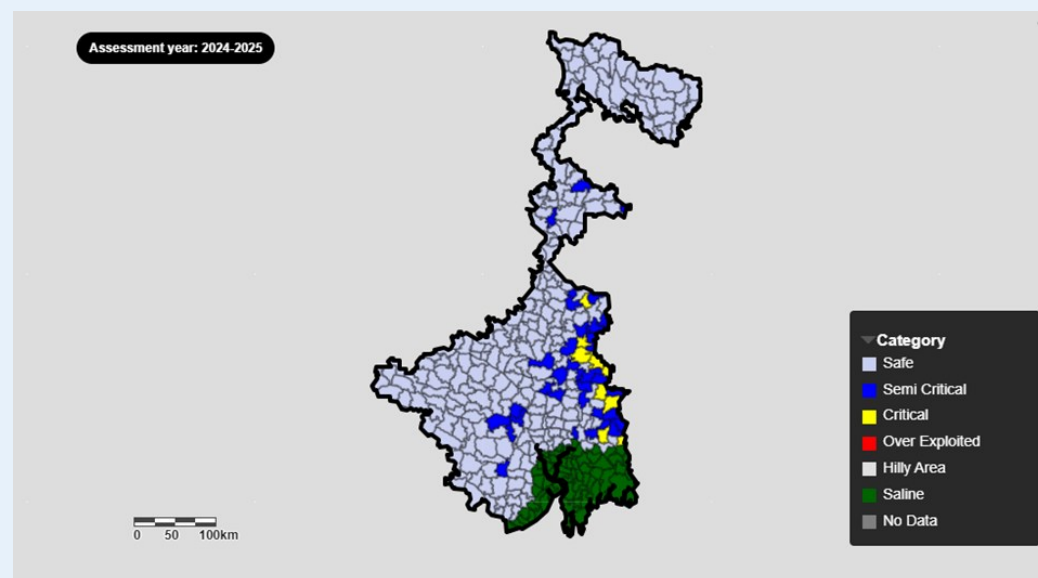
<https://ingres.iith.ac.in>

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Central Ground Water Board
Department of Water Resources, RD & GR
Ministry of Jal Shakti, Government of India



**Dynamic Ground Water Resources, Ground Water Level &
Ground Water Quality of West Bengal, 2025**

November, 2025

Background

- ◆ Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- ◆ Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- ◆ As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Block).
- ◆ The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- ◆ GWRA- 2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad.

Salient Features

1	Rainfall	1,737.85 mm
2	Hydrogeology	Nearly two third area of the state is occupied by unconsolidated sediments; the western part of the state is partly occupied by the hard rocks.
3	Recharge Worthy Area of the State	79.88 Thousand Sq. Km
4	Assessment Unit (AU) Type / Number	Block / 345 Numbers
5	Average area of Assessment Unit	231.54 Sq. Km

Findings

	Attribute	GWRA-2017	GWRA-2020	GWRA-2022	GWRA-2023	GWRA-2024	GWRA-2025
1	Total Annual Ground Water Recharge (in bcm)	29.33	29.33	23.61	26.29	25.89	25.85
2	Annual Extractable Ground	26.56	26.56	21.42	23.9	23.56	23.50
3	Annual Ground Water Extraction (in bcm)	11.84	11.84	10.07	10.71	10.75	10.62
4	Stage of Ground Water Extraction (in %)	44.59	44.60	47.01	44.81	45.63	45.19

bcm: Billion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

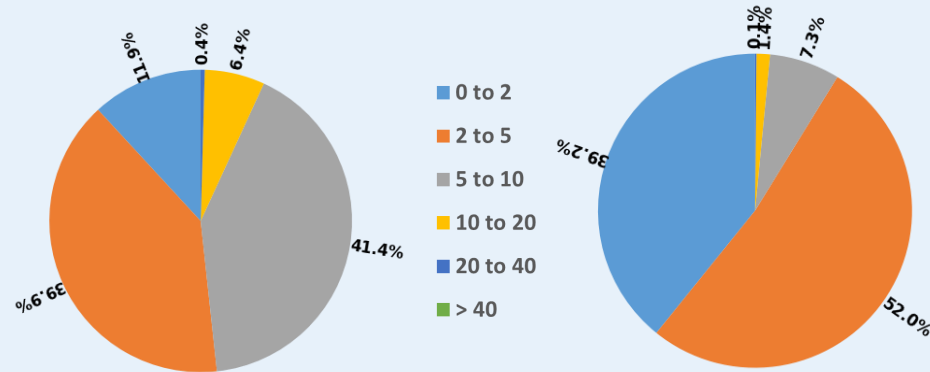
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs
1	Safe	191	71	19321	71.27	232	67.24	241	69.86	239	69.28	242	70.14
2	Semi-critical	76	28	76	28.36	31	8.98	32	9.28	36	10.43	34	9.86
3	Critical	1	0.37	1	0.37	22	6.37	12	3.48	10	2.9	9	2.61
4	Over-exploited												
5	Saline					60	17.39	60	17.39	60	17.39	60	17.39
Total number of AUs		268		268		345		345		345		345	

Recommendations: -

- * Nearly two third area of the state is occupied by unconsolidated sediments; the western part of the state is partly occupied by the hard rocks.
- * The ground water resource assessment (in 2024) for the State of West Bengal has been carried out as per GEC 2015 guidelines through 'IN-GRES', with blocks as primary assessment units. All 344 blocks of the State of West Bengal and one (01) urban area as Kolkata Municipal Corporation is assessed. Total annual groundwater recharge has been estimated at 25.85 bcm, and the annual extractable groundwater resource has been estimated at 23.50 bcm. The current annual groundwater extraction for all uses has been estimated at 10.62 bcm, which translates into a stage of groundwater extraction of 45.19%. Out of 345 assessment units in the State, 9 units (2.61%) have been categorized as 'Critical', 34 units (9.86%) as 'Semi-Critical', 242 units (70.14%) as 'Safe', and 60 units (17.39%) as 'Saline'. There is no 'Over-Exploited' block in the State.
- * In the safe category areas of West Bengal, State Government can judiciously develop the ground water resource mainly for agricultural use, however, at no point of time the extraction level should exceed 70%.
- * National Aquifer Mapping & Management Programme (NAQUIM) Reports prepared by CGWB (<https://cgwb.gov.in/cgwbpm/>) which are also being shared with State/District Authorities and Ground Water Year Book published by CGWB having water level & water quality data may be used in Ground water management. (<https://cgwb.gov.in/cgwbpm/>).
- * State may review their free/subsidized electricity policy to farmers (if applicable), bring suitable water pricing policy and may work further towards crop rotation/diversification/other initiatives to reduce overdependence on groundwater.
- * Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in West Bengal

1. Depth to Groundwater Level

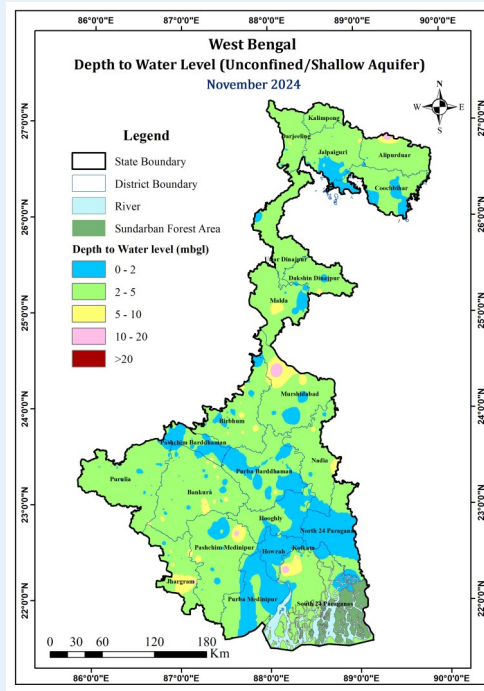
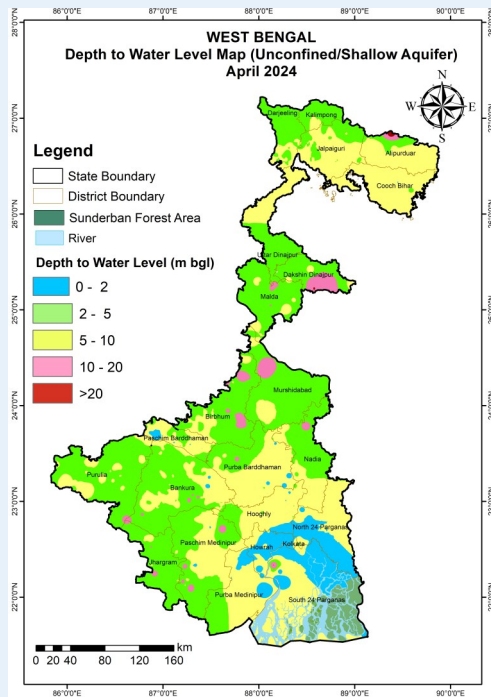


Pre-Monsoon (May 2024)

About 12% of wells recorded groundwater levels in 0–2 m bgl, 40% in 2–5 m bgl, 41% in 5–10 m bgl, 6% in 10–20 m bgl, and a few wells (>0%) beyond 20 m bgl, with overall levels ranging from 0.3 m to 26.22 m bgl, indicating generally moderate groundwater with deeper pockets in hard rock and older alluvial regions.

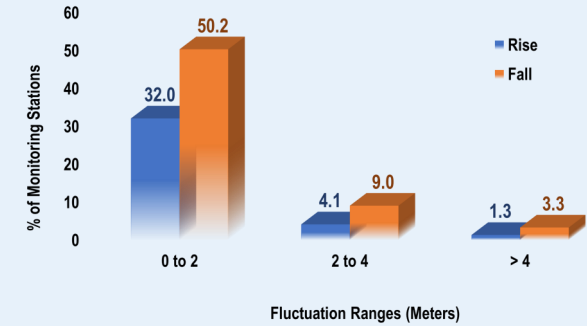
Post-Monsoon (November 2024)

About 39% of wells recorded groundwater levels in 0–2 m bgl, 52% in 2–5 m bgl, 7% in 5–10 m bgl, and 1% in 10–20 m bgl, with overall levels ranging from 0.06 m to 20.48 m bgl, indicating generally shallow groundwater with isolated deeper pockets.



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

